

MOHAN SINGH

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[LinkedIn](#) [GitHub](#) [Portfolio](#)

SUMMARY

BCA graduate (2025) with hands-on experience building and testing REST APIs using Python, Flask, and SQLAlchemy. Developed a JWT-authenticated Expense Tracker API with 28 pytest tests and performed exploratory data analysis on a 9,994-row retail dataset. Comfortable with SQL databases, Git/GitHub, and deploying Python applications.

EDUCATION

Noida International University <i>Bachelor of Computer Applications (BCA)</i>	2022 – 2025
Kendriya Vidyalaya, Noida <i>Senior Secondary (12th)</i>	2022
Kendriya Vidyalaya, Noida <i>High School (10th)</i>	2020

SKILLS

Languages: Python, JavaScript, C

Frameworks & Libraries: Flask, SQLAlchemy, Flask-JWT-Extended, WTForms, Pandas, Matplotlib, Seaborn, Plotly, Tkinter

Web Technologies: HTML5, CSS3, Bootstrap 5, REST API, JSON

Databases: MySQL, PostgreSQL, SQLite

Developer Tools: Git, GitHub, Postman, pytest, VS Code

Concepts: OOP, JWT Authentication, Data Analysis, Machine Learning Fundamentals

PROJECTS

Expense Tracker REST API [GitHub](#)

Python, Flask, SQLAlchemy, JWT, bcrypt, pytest

- * Built a REST API with JWT authentication, bcrypt password hashing, and per-user data isolation across all endpoints
- * Implemented soft delete, pagination, and Decimal type for exact monetary arithmetic — avoiding float rounding errors
- * Added category-based budget limits with warning and exceeded alerts returned in monthly summary response
- * Separated concerns into models, services, validators, and routes layers — routes contain zero business logic
- * Wrote 28 pytest tests covering auth, CRUD, edge cases, data isolation, and budget alerts

Superstore Sales Analysis [GitHub](#)

Python, Pandas, Matplotlib, Seaborn, Plotly

- * Performed EDA on a 9,994-row retail dataset — cleaned missing values, removed duplicates, and extracted business insights
- * Built 4 visualizations including an interactive sales vs profit scatter chart using Plotly
- * Identified Tables sub-category operating at a loss (-\$17,725) despite high sales volume
- * Exported structured summary report to CSV covering sales, profit, and order count by category

Cafe Finder REST API [GitHub](#)

Python, Flask, SQLAlchemy, SQLite

- * Built a REST API with Flask exposing 6 endpoints covering all HTTP methods (GET, POST, PATCH, DELETE)
 - * Designed a SQLite database using SQLAlchemy ORM with a Cafe model storing 10+ attributes including location and amenities
 - * Implemented API key authentication on the DELETE endpoint with correct HTTP status codes (200, 403, 404) across all routes
 - * Serialized database objects to JSON using a custom `to_dict()` method for clean, consistent API responses
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CERTIFICATIONS

- * Complete Python Pro Bootcamp — Udemy
- * Samsung Innovation Campus (SIC) — Artificial Intelligence
- * TCA Training & Development — Neural Networks & TensorFlow
- * Ducat Industrial Visit — Data Analytics